CP–7922

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

Part II – ENGLISH – PROSE AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the essential of education?

2. Who was called as an Iron Lady of India?

3. What did happen when Scott removed the boy's button?

4. Why did the boy make the mistake purposefully in *Crime and Punishment*?

5. How did Atwood call survival?

6. What is patriotism according to Sarojini Naidu?

7. (a) Life (b) baby (Change the words into Plurals)

8. Pick out the adverbs in the following sentences :

   (a) The boy walked slowly.

   (b) You must stay here.

9. He bought a pen last week. (Change into Negative)

10. Murugan lives ———— America. (Fill in the blank with suitable preposition).

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What, according to Livingstone, is the attitude of an educated man to the problems of his life?

Or

(b) What are the powers the modern youth have according to India Gandhi? Describe.

12. (a) Write about the causes of habits.

Or

(b) Why does the teacher support the boy at the end of the story *Crime and Punishment*? Explain.

13. (a) How did Margret criticise Canadian literature in her Survival.

Or

(b) Elucidate the journey of Sarojini Naidu as a freedom fighter.

14. (a) Pick out the Adjectives in the following sentences :

(i) Some water is left at the bottom of the tank.

(ii) There is enough food for two.

(iii) I have two sisters.

(iv) Which book did you buy?

(v) This boy is a good cricketer.

Or

(b) Fill in the blanks with suitable tense forms :

(i) We ———— our old house last week; we ———— a new house very soon. (sell/buy)

(ii) Leela ———— music for over a year. (learn)

(iii) They ———— to break into a house when the police ———— them. (try/arrest)

15. (a) Fill in the blanks with suitable Articles :

(i) Mr. Shankar is ———— M.P.

(ii) I love ———— music very much.

(iii) ———— rainbow is ———— thing of beauty.

(iv) She is playing ———— piano.

Or

(b) Fill in the blanks by using may, might, can or could :

(i) Even as a child I ———— swim across this river.

(ii) ———— you live long!

(iii) You ———— use my pen.

(iv) If she decides, who ———— can stop her?

(v) We were worried that you ———— get lost.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Write a short note on aim of education.

Or

(b) Write an essay on Indira Gandhi's views on Youth.

17. (a) Justify the title of the story *Crime and Punishment*.

Or

(b) Describe the theme of *The Vision of Patriotism*.

18. (a) Fill in the blanks with suitable prepositions :

(i) Vimala goes ———— school every morning ———— nine.

(ii) Indra was punished ———— her teacher ———— copying ———— her neighbour's book.

(iii) I begin my day ———— a cup of coffee prepared ———— my mother.

(iv) He goes ———— bed ———— ten o'clock and gets up ———— sunrise.

Or

(b) Fill in the blanks with correct tense form :

(i) She ———— him when he ———— tomorrow. (receive / come)

(ii) He ———— as long as you ———— (wait/want)

(iii) Unless you ———— the fees, they ———— you out of the class. (pay / send)

(iv) If you ———— away from the meeting, they ———— you in future. (stay / ignore)

(v) We ———— here for ten years by the end of this year. (live)

(vi) The children are ———— before I reach home. (go to bed)

———————

CP–7923	Sub. Code
	12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

GENERAL ANATOMY AND PHYSIOLOGY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Pancreas.

2. Urinary bladder.

3. Lymph nodes.

4. Blood pressure.

5. Cerebrospinal fluid.

6. Bronchi.

7. Involuntary muscles.

8. Bile Juices.

9. Coagulation of blood.

10. What is Goiter?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Structure of skin.

Or

(b) Structure and functions of ureter.

12. (a) Functions of WBCs.

Or

(b) Immunoglobulins.

13. (a) Functions of bile.

Or

(b) Cardiac cycle.

14. (a) Anatomy of brain.

Or

(b) Functions of digestive system.

15. (a) Types and classifications of joints.

Or

(b) Structure of skeletal muscle.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Heart – position, relations, chambers, flow of blood through heart and coronary circulation.

Or

(b) Describe the anatomy of Pancreas.

17. (a) Write anatomy of lungs and describe the mechanism of respiration.

Or

(b) Write about menstrual cycle.

18. (a) Anatomy of adrenal gland.

Or

(b) What are the pituitary hormones? Elaborate on growth hormone.

CP–7924

Sub. Code
13

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

GENERAL AND OCULAR BIOCHEMISTRY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Diabetics mellitus.

2. Zwitter ions.

3. Hydrolysation.

4. Aliphatic amino acids.

5. Example for polysaccharide.

6. Xeropthalmia.

7. Types of immunoglobulin.

8. Cellulose.

9. Define electrolytes.

10. Mutarotation.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write about fat soluble vitamin.

Or

(b) Explain about mechanism of enzyme.

12. (a) Write about atherosclerosis.

Or

(b) Write short notes on lens.

13. (a) PUFA.

Or

(b) Role of retinal vision.

14. (a) Write in detail about blood glucose.

Or

(b) Write about biochemical composition of tear film.

15. (a) Explain about haemoglobin.

Or

(b) Estimate the blood creatinin.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write about Glycolysis.

Or

(b) Discuss in detail about general properties of protein.

17. (a) Sources, function, deficiency of water soluble vitamins.

Or

(b) Classification of carbohydrates and explain about its physical properties.

18. (a) Fatty acid biosynthesis.

Or

(b) Michaelis-Menten Hypothesis.

———————

CP–7925

Sub. Code
14

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

GEOMETRICAL OPTICS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define convergence of wave.

2. Give the Fermat's principle.

3. What is chromatic aberration?

4. Define wave length.

5. What is optical path length?

6. Define speed.

7. Define focal points.

8. Define light.

9. Give the refractive index of a medium.

10. Give the Law of refraction.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What is coma in lens aberration?

Or

(b) Describe geometrical and optical path length of rays.

12. (a) Different types of lenses.

Or

(b) Explain lateral and axial magnification.

13. (a) Total internal reflection.

Or

(b) Dual nature of light.

14. (a) Explain principle points and principle planes.

Or

(b) Theory of optical fibres.

15. (a) Spherical aberration.

Or

(b) Derive Vergence equation.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain refraction through a concave surface.

Or

(b) Explain in detail the geometrical theory of light propagation in optical fibers.

17. (a) Explain the following in detail :

(i) Astigmatism,

(ii) Distortion.

Or

(b) Derive the law of reflection based on fermat's principle.

18. (a) Dispersion of a prism.

Or

(b) Derive the relation between U,V,R for refraction at concave surface.

CP–7926

Sub. Code
15

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

NUTRITION

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Name two disaccharides.

2. Define minerals.

3. What are body building foods?

4. Name two dietary sources of Vitamin A.

5. Name the essential fatty acids.

6. What are trace elements? Give two examples.

7. What is gross energy value of food?

8. Define Basal Metabolic Rate (BMR).

9. What is scurvy and rickets?

10. Define Hyperlipedemia.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Menu planning.

Or

(b) Recommended dietary allowances.

12. (a) Classify lipids.

Or

(b) Calcium deficiency and excess.

13. (a) Fat soluble vitamins.

Or

(b) Iron metabolism.

14. (a) Supplementary food.

Or

(b) Egg and its nutritive value.

15. (a) Milk and Milk products.

Or

(b) Food Pyramid.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Protein energy malnutrition.

Or

(b) Vitamin A deficiency.

17. (a) Diet planning for preschool children.

Or

(b) Assessment of nutritional status.

18. (a) Types of anemia or its nutritive management.

Or

(b) Carbohydrates and its nutritive values.

CP–7927

Sub. Code
16

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

COMPUTERS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give two types of main memory.

2. What is the function of ALU?

3. Convert the decimal number 31.572 into octal.

4. What is binary coded decimal?

5. What is an operating system?

6. How will you maximize or minimize the window?

7. List down the various formats that you can apply on the text data entered in a cell.

8. What are the elements of a slide?

9. What is a browser? Give an example.

10. What is the use of anti virus?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Where do you use computer in day to day life? Explain briefly.

Or

(b) Describe the different types of printers.

12. (a) Explain binary number system.

Or

(b) Perform the following operations:

(i) $10101 \cdot 10101 - 1101 \cdot 011$

(ii) $(11011) \div (101)$.

13. (a) How can you create and rename a folder?

Or

(b) Write a note on "My computer".

14. (a) Describe the different ways to edit a cell.

Or

(b) Give a procedure to set times in a slide.

15. (a) What is Internet? Also compare with intranet.

Or

(b) How can you prevent virus?

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Describe the different types of computers.

Or

(b) What is hexadecimal number system? How can you convert it into decimal? Give an example.

17. (a) Describe the uses of any four accessories.

Or

(b) Explain the concept of mail merge with an example.

18. (a) Explain the structure of an email address. Also explain how can you compose and forward a mail.

Or

(b) Explain the activities that you can do on the Internet.

———————

CP–7928

Sub. Code
21

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

Part II — English

PROSE, EXTENSIVE READING AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Who are the three men that Kalam worked with?

2. What is the essence of the scientific point of view?

3. 'Gandhi was not of the common clay' – Explain.

4. Who argued in the nursery?

5. What are the three types of Discipline according to John Holt?

6. Who is Bob?

7. What happened to Guru Nayak?

8. What did the conjurer take at first?

9. Give an example for Adverbial Clause of Time.

10. Change the Tense in Reported Speech – Shyam said that he had bought a car.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) On what grounds does Kalam criticise Indians? Describe.

Or

(b) What are the forces that prevent us from stamping out diseases?

12. (a) Write a paragraph on the opening of Nehru's Speech.

Or

(b) What does Robert Lynd think of arguing in schools and colleges?

13. (a) Briefly describe the tricks used by the Conjurer.

Or

(b) Comment on the theme of The Four Brothers.

14. (a) Analyse Adverbial Clause of Manner with some examples.

Or

(b) Describe Relative Pronoun with few examples.

15. (a) Change the following sentences into other degrees :

(i) No other girl is so good as Rani

(ii) Somu is more active than any other boy.

(iii) Chennai is the largest city in Tamil Nadu.

Or

(b) Change the following sentences into possible speech :

(i) "Goodbye, Ram!" he said

(ii) He requested me to have dinner with her

(iii) He said, "Let's go for lunch"

(iv) She told Sita to sit down

(v) He asked when they had come.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write an essay on *My Visions For India*.

Or

(b) How can a scientific attitude help man? Illustrate.

17. (a) What is a message delivered by Nehru? Describe.

Or

(b) Attempt a note on Discipline of Nature.

18. (a) Illustrate kinds of Adjectives with examples.

Or

(b) Fill in the blanks with suitable conjunctions :

(i) Run fast, ———————— you cannot catch the bus.

(ii) ———————— you work hard, you can pass.

(iii) They failed ———————— they were lazy.

(iv) Rama was angry ———————— he kept silent.

(v) ———————— you work hard, you can't pass.

———————

CP–7929

Sub. Code
22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

OCULAR ANATOMY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Structures in the superior orbital fissure.

2. Plica semilunaris.

3. Meibomian gland.

4. Layers of cornea.

5. Write about macula.

6. LPS muscle.

7. Contents of orbit.

8. Parts of uvea with nerve supply.

9. Extra occular muscle.

10. Conjunctiva.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) The optic-chiasma anatomy and arrangement of nerve fibres.

Or

(b) The orbicularis oculi muscle.

12. (a) Describe the anatomy of the eyelid.

Or

(b) Anatomy of vitreous.

13. (a) Write in detail about orbital anatomy.

Or

(b) Explain the anatomy of uvea.

14. (a) Anatomy of crystalline lens.

Or

(b) Anatomy of angle of anterior chamber.

15. (a) Sclera.

Or

(b) Anatomy of optic nerve

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Draw a diagram of visual pathway. Explain about the course of the visual fibres.

Or

(b) Draw the longitudinal section of the eyeball and mark the parts.

17. (a) Write in detail about the bones forming the base, roof, medial and lateral wall of the orbit.

Or

(b) Write in detail about anatomy of cornea.

18. (a) Write on the origin, course and insertion of superior oblique muscle.

Or

(b) Describe the anatomy of the extraocular muscles and their actions.

CP–7930

Sub. Code
23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

OCULAR PHYSIOLOGY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Fick's axis

2. Steriopsis

3. Ocular changes during accommodation

4. Functions of cornea

5. Define contrast sensitivity

6. Anomalies of binocular single vision

7. Factors affecting visual acuity

8. Pannum's area

9. Versions

10. Functions of visual cortex.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Presbyopia.

Or

(b) Measurement of contrast sensitivity.

12. (a) Visual evoked potential.

Or

(b) Theories of accommodation.

13. (a) Physiology of aqueous humour.

Or

(b) Pupillary reflex.

14. (a) Herrings law and Sherington's law. Explain with example.

Or

(b) Test for binocular single vision.

15. (a) Visual cycle.

Or

(b) What is an electro Retinogram? Write about its components and uses.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Mention 5 layers of cornea and explain factors responsible for corneal transparency.

Or

(b) Draw a diagram of Lacrimal apparatus and drainage system.

17. (a) Colour vision, mention theories of colour vision and write a notes on colour vision defect.

Or

(b) Visual acuity and papillary reflexes.

18. (a) Types and measurement of contrast sensitivity.

Or

(b) Discuss in detail about binocular single vision.

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	24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

PHYSICAL OPTICS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define wave velocity.

2. State the laws of refraction at a spherical surface.

3. What is coherence?

4. Define visibility of fringes.

5. What is a phase grating?

6. State Rayleigh's criterion.

7. Define Degree of polarization.

8. What is a hologram?

9. Define stimulated emission.

10. Give the wavelength range of visible spectrum.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Discuss the superposition of two simple harmonic waves at right angles to each other.

Or

(b) State and explain the laws of reflection at a plane surface.

12. (a) What is meant by interference of light? State the conditions for the production of interference fringes.

Or

(b) Give a brief account of thin film antireflection coatings.

13. (a) Write a note an amplitude grating.

Or

(b) Explain the dispersive power of a grating.

14. (a) Give an account of quarter wave plate.

Or

(b) Write a note on Wollaston Prism.

15. (a) Give a brief account of Infrared spectrum.

Or

(b) Discuss the resolution of binoculars.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) State and explain Huygeis principle. Explain the phenomenon of reflection and refraction of a plane wave on a spherical surface on the basis of wave nature of light.

Or

(b) Describe Newton's ring experiment to determine the wavelength of a monochromatic source of light with necessary theory.

17. (a) Give the theory of zone plate.

Or

(b) Explain the Holography. Describe an experimental arrangement for recording and reconstruction of hologram.

18. (a) Discuss briefly the various sources of spectrum.

Or

(b) Write a note on :

(i) Veal prism

(ii) Interference in Wedge-Shaped films.

CP–7932

Sub. Code
25

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

MICROBIOLOGY AND PATHOLOGY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define microbiology with two ocular applications.

2. What is moist heat sterilization?

3. Give any two applications of acid fast staining.

4. What is an antibiotic give an example?

5. Give two treatments for staphylococcal blepharitis.

6. What is a tumor? Give two examples.

7. Define tissue injury.

8. What is a causative organism of leprosy and tuberculosis?

9. Give two examples for immune mediated reactions.

10. Mention the types of cataract.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the steps involved in the fixing of the slides?

Or

(b) What is virology? Give the clinical importance of pox and retro virus.

12. (a) Write a short on spirochetes infection diagnosis and treatment.

Or

(b) What are the steps involved in the healing and repair mechanism during a tissue injury?

13. (a) Brief about toxocara and filaria species.

Or

(b) Write a short note on Retinoblastoma.

14. (a) What is a normal ocular flora?

Or

(b) Explain about Hordeolum and its types.

15. (a) Elaborate about specimen collection from a corneal ulcer and vitreoustaping.

Or

(b) Brief on Giemsa staining.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Define culture and different types of culture used in the field of microbiology.

Or

(b) Explain about conjunctivitis and its types.

17. (a) Write a detailed note on :

(i) Squamous all

(ii) Carcinoma

(iii) Malignant melanoma.

Or

(b) Enumerate the disease manifestation, diagnosis and treatment of fusarium, mucor and candida species.

18. (a) What is a hypersensitivity reactions, brief about its types?

Or

(b) Brief about the clinical importance, diagnosis and treatment of streptococci and pneumococci species.

———————

CP–7933

Sub. Code
26

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Optometry

ENVIRONMENTAL STUDIES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Atmosphere.

2. Green house.

3. Afforestation.

4. Abiotic factors.

5. Renewable energy.

6. Mineral cycling.

7. Biomass.

8. Decomposers.

9. Ethical value.

10. Acid rain.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the causes influence the deforestation? Explain.

Or

(b) How do you conserve the fresh water? Explain.

12. (a) Give an account on usage of mineral resources

Or

(b) Write the consequences of unscientific construction practice.

13. (a) Briefly explain the food chain in various ecosystem.

Or

(b) Write a note on indirect value of biological resources.

14. (a) What are the major threats to biodiversity? Explain.

Or

(b) Give an note on endangered species of India.

15. (a) Write the methods of control and prevention of water pollution.

Or

(b) Write an essay on noise pollution.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Explain the importance of public awareness on environment.

Or

(b) Describe the role of an individual in conservation of natural resources.

17. (a) Write a detailed note on the structure and functions of Pond ecosystem.

Or

(b) Discuss India as a mega Biodiversity.

18. (a) Give an account on nuclear hazards on human and environment.

Or

(b) What are the major causes for the marine pollution? Explain in detail.

———————

CP–7934

Sub. Code
31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Communication.

2. Name the kinds of verbal and non-verbal Communication.

3. What is meant by Phrasal Verb?

4. Differentiate homonyms and homophones.

5. List out some common mistakes in English Usage.

6. Cite some examples for misplaced words.

7. Explain Intonation.

8. What do you mean by stress?

9. Name the types of reading.

10. What are the characteristics of good handwriting?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write a short note on the ways of challenging the barriers of communication.

Or

(b) Explain the types of communication.

12. (a) Define unnecessary words with examples.

Or

(b) Write a brief note on verbs often confused.

13. (a) What is a Relative Clause? Explain.

Or

(b) Define Transformation of Sentences with - Examples.

14. (a) Explain sounds.

Or

(b) Define intonation and its types.

15. (a) Importance of communication- explain.

Or

(b) Define SQ3R.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write essay on barriers to Communication.

Or

(b) Detail how vowels and consonants are important.

17. (a) Elaborate Active and Passive Voice.

Or

(b) Write an essay on confused words with examples.

18. (a) Primary and Secondary Stress- Discuss.

Or

(b) Discuss in detail the Mechanics of Writing.

———————

CP–7935

Sub. Code
32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

VISUAL OPTICS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Donder's rule of hypermetropia.

2. AC/A ratio.

3. Anisometropia.

4. Back vertex distance.

5. Define Myopia.

6. Astigmatic fan.

7. Prism used in Ophthalmology.

8. Disadvantages of aphakic glasses.

9. Define colour vision.

10. Define contrast sensitivity.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Optical aberrations of the eye.

Or

(b) Optics of Astigmatism.

12. (a) Jackson Cross Cylinder (JCC).

Or

(b) Fogging Technique.

13. (a) Duochrome Test.

Or

(b) Components of optical system.

14. (a) Uses of prisms in Ophthalmology.

Or

(b) Define Amblyopia and its types.

15. (a) Factors affecting visual acuity.

Or

(b) Anomalies of accommodation.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write in detail about keratometry.

Or

(b) Principles and techniques of retinoscopy.

17. (a) Write in detail about reduced eye.

Or

(b) Optics, types, clinical features and management of Hypermetropia.

18. (a) Define visual acuity. Discuss in detail the various types of visual acuity charts and their applications .

Or

(b) Subjective methods of verification of refraction.

CP–7936

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

OCULAR DISEASES – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Stye.

2. Nanopthalmos.

3. Congenital cloudy cornea.

4. Mention the signs of Bacterial Keratitis.

5. What is an AC IOL?

6. Define 'Dry Eye'.

7. Explain the pathophysiology of allergic conjunctivitis.

8. What is iridotomy and when is it done?

9. Explain congenital lamellar cataract.

10. Clinical features of Staphylococcal blepharitis.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Preseptal cellulitis.

Or

(b) Pterygium.

12. (a) Vernal Keratoconjunctivitis.

Or

(b) Viral Keratitis.

13. (a) Primary angle closure glaucoma.

Or

(b) Vitamin A deficiency.

14. (a) Fill in the blanks :

(i) ———————— is an infection of the nasolacrimai sac, frequently caused by nasolacrimal duct obstruction.

(ii) ———————— is characterized by discrete nodular areas of corneal or conjunctival inflammation.

(iii) ———————— is a rare, noninflammatory corneal ectasia characterized by diffuse protrusion and thinning of the cornea.

(iv) ———————— organism causes Trachoma.

(v) ———————— nodules are seen on the surface of the iris in Anterior Uveitis.

Or

(b) Match the following :

(i)	Viral Keratitis	(1)	Fuch's dystrophy
(ii)	Descemet's membrane and Endothelium	(2)	Hyaline deposits
(iii)	Epithelium	(3)	Herpes Simplex
(iv)	Granular dystrophy	(4)	Lattice dystrophy
(v)	Stroma	(5)	Map-dot fingerprint dystrophy

15. (a) Pellucid Marginal Degeneration.

Or

(b) Meesman corneal dystrophy.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Describe the types and clinical features of Ptosis. Write about the clinical evaluation of Ptosis.

Or

(b) Describe the pathophysiology and the signs of Grave's Eye Disease.

17. (a) Describe the clinical features and management of Anterior Uveitis.

Or

(b) Describe the clinical features and management of Secondary glaucoma.

18. (a) Describe the clinical features, signs and symptoms of the types of senile cataract.

Or

(b) Write about Trabeculectomy.

CP–7937

Sub. Code
34

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

OPTOMETRIC INSTRUMENTS – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Maddox rod.

2. Pinhole.

3. Define LASER.

4. Two advantages of spot retinoscopy.

5. What is OCT?

6. Name two charts used for illiterate patients.

7. Write two advantages of Schiotz tonometer.

8. What is principle of auto ref rafractometer?

9. Mention two uses of ultrasound in ophthalmology.

10. Name two instruments used to measure squint.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Trial Set.

Or

(b) Direct ophthalmoscope.

12. (a) Color vision test.

Or

(b) RAF ruler.

13. (a) A scan biometry.

Or

(b) Logmar chart.

14. (a) Keratometer.

Or

(b) Tests for binocular single vision.

15. (a) Slitlamp.

Or

(b) Amsler grid.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Retinoscope.

Or

(b) Visual acuity charts for children.

17. (a) Tonometer.

Or

(b) Laser.

18. (a) Instruments to measure visual fields.

Or

(b) Ophthalmic photography.

CP–7938

Sub. Code
35

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

GENERAL AND OCULAR PHARMACOLOGY

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is detoxication?

2. Define Bio availability.

3. Write the uses of following drugs
   (a) Adrenaline
   (b) Alpha methyl depa.

4. What is SAR with examples?

5. What is the use of anaesthetic drugs?

6. Dale's vasomotor reversal.

7. What is microsome?

8. Write four tear substitutes.

9. What are the drugs used in CNS stimulant?

10. What is insomnia and how to treat it?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write nature and sources of drugs.

Or

(b) Factors influencing the absorption and metabolism of drugs.

12. (a) Write some classification of drugs.

Or

(b) Mechanism of drug action.

13. (a) Write the pharmacological action and uses of general anaesthetics.

Or

(b) What is receptor? Write different types of receptor.

14. (a) Narcotic analgesics.

Or

(b) Local anaesthetics.

15. (a) General considerations of ANS.

Or

(b) Viscoelastic substances.

Part C

(3 × 10 = 30)

Answer **all** questions.

16. (a) Routes of drug administration in eye.

Or

(b) Explain the adverse drugs reactions in man and manifestations of ADR.

17. (a) What is NSAID and explain it.

Or

(b) Write briefly, parasympathomimetic drugs.

18. (a) What is ocular diagnostic drugs and explain it.

Or

(b) What is glaucoma and write antiglaucoma drugs?

CP–7939

Sub. Code
36

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Optometry

CLINICAL EXAMINATION OF VISUAL SYSTEM

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. RAPD.

2. Krimsly test.

3. TNO cards.

4. Anomalies of colour vision.

5. Name of few perimeter.

6. Occlusion therapy.

7. Amsler test.

8. Principles of pinhole.

9. Schirmer's test.

10. Smooth muscle.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe history taking.

Or

(b) Prism bar cover test.

12. (a) Difference between direct and indirect ophthalmoscopy.

Or

(b) Procedure of confrontation test.

13. (a) Jackson cross cylinder.

Or

(b) Binocular balancing.

14. (a) Different methods of dynamic retinoscope.

Or

(b) Drugs used for cycloplegic refraction.

15. (a) Principles and procedure of Applanation Tonometry.

Or

(b) Measurement of inter papillary distance.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Principle, uses, merits and demerits of autorefractor.

Or

(b) Discuss the technique, Principles and practice of retinoscopy.

17. (a) Grades and anomalies of binocular single vision.

Or

(b) Tear film tests.

18. (a) Examination of pupilary reflex.

Or

(b) Write in detail about various techniques used to measure the intra ocular pressure.

———————